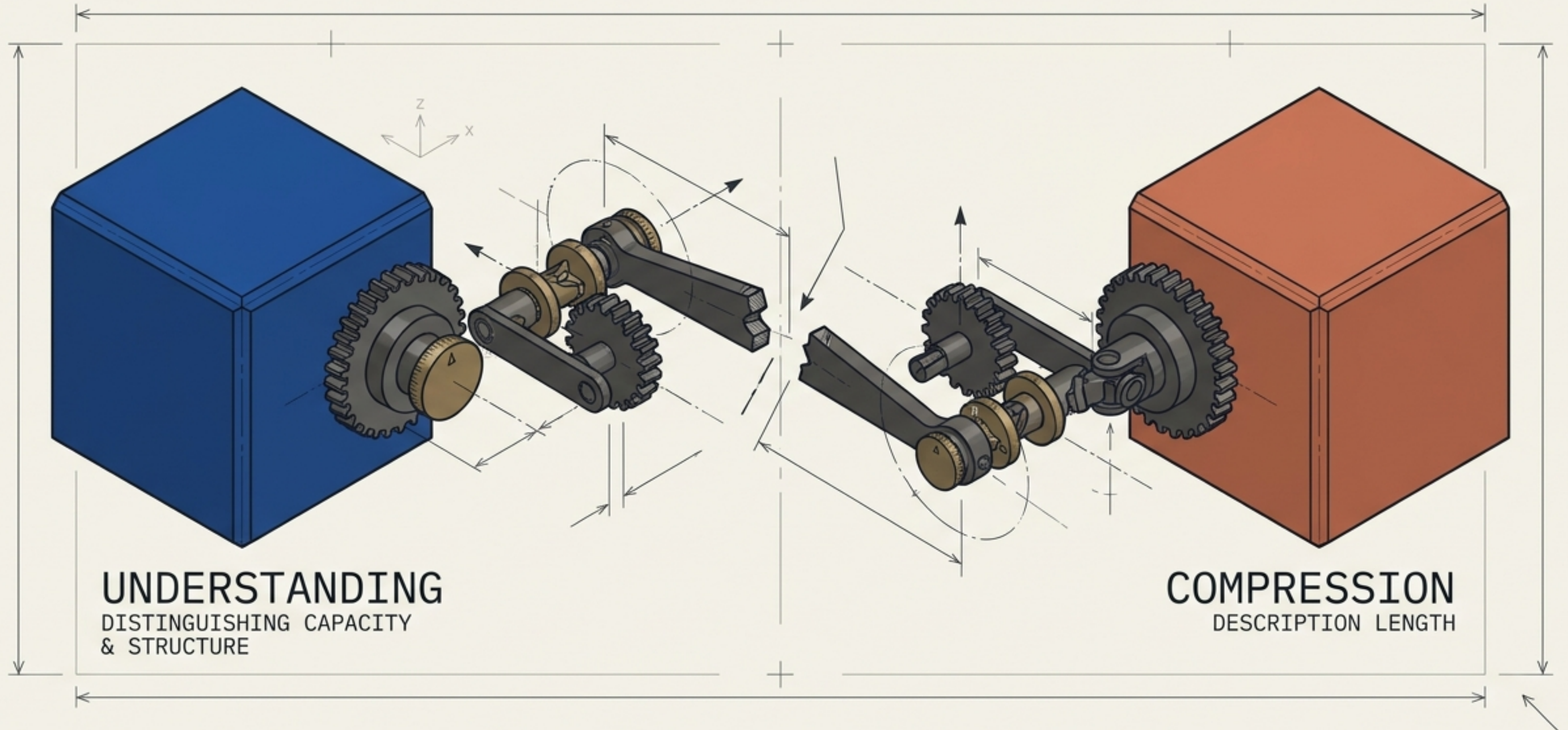
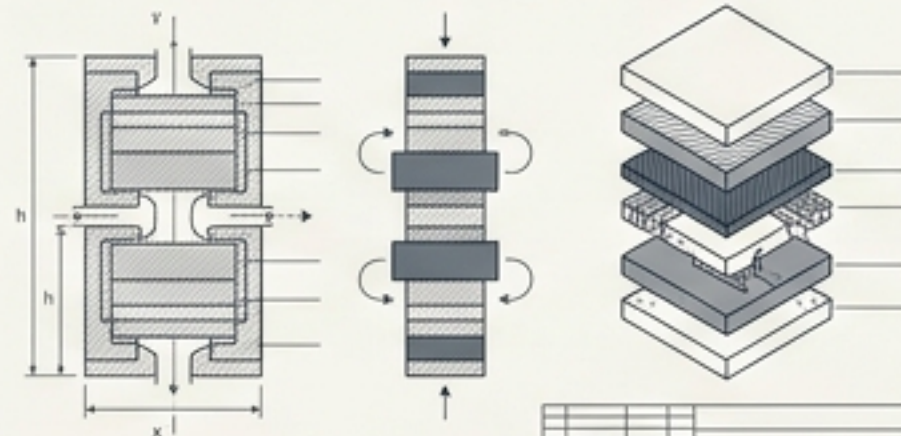


The Compression Fallacy

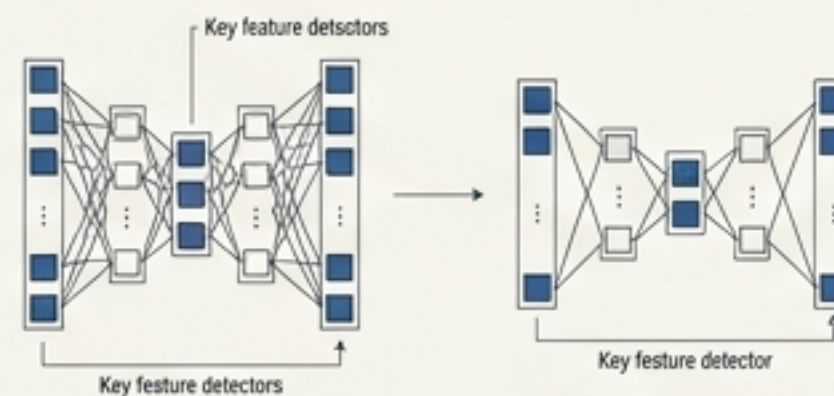
Why shorter is not the same as better understood.



Schmidhuber's Compression Progress



Sparse Autoencoder Interpretability



Minimum Description Length (MDL)



THE ASSUMPTION:

“The amount of information in a pattern of bits equals the length of the smallest computer program capable of generating those bits.” — Daniel Hillis

A fundamental law of physics

THE COROLLARY WE ACT ON:

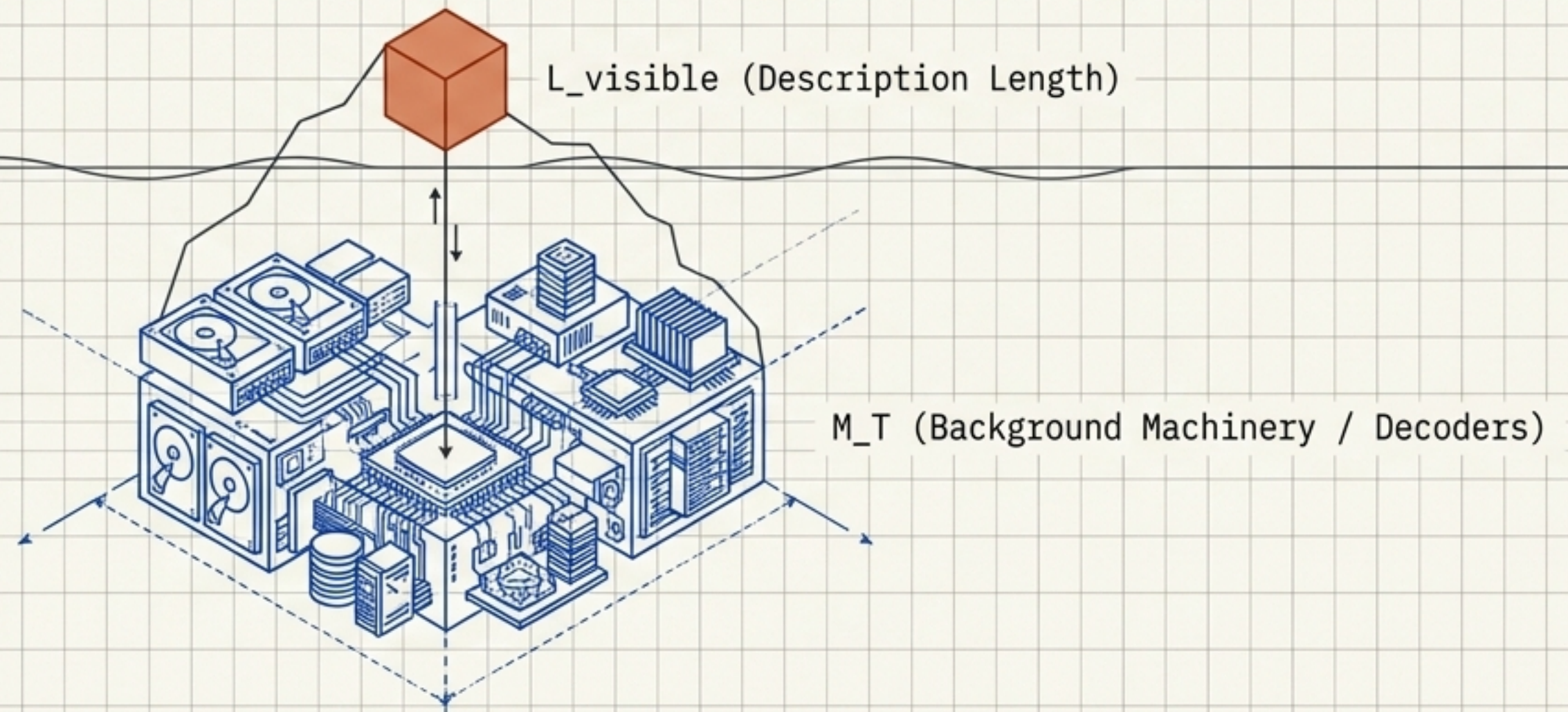
“A system that produces a shorter description of a domain has understood that domain better. We treat shortness as the ultimate proxy for true structural insight.”

Proposition 1: The Illusion of Eliminated Complexity

Reducing per-object description length does not eliminate complexity; it merely displaces it into the background machinery.

- A 3-character shell command invokes an entire operating system.
- A 128-dimensional latent vector invokes a billion-parameter decoder.

Reporting length without reporting the presupposed machinery is measuring an incomplete equation.

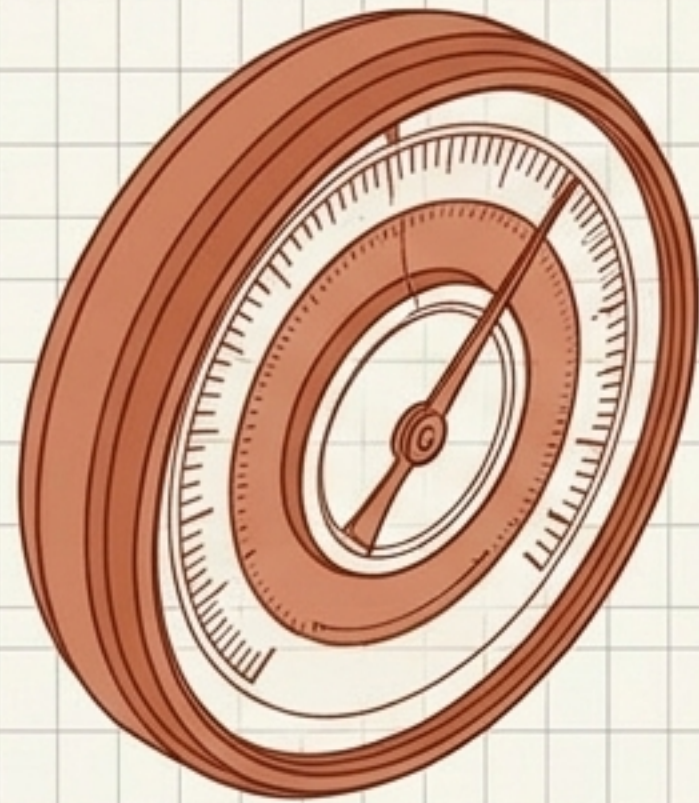


Proposition 2: Breaking the Gears of Non-Identification

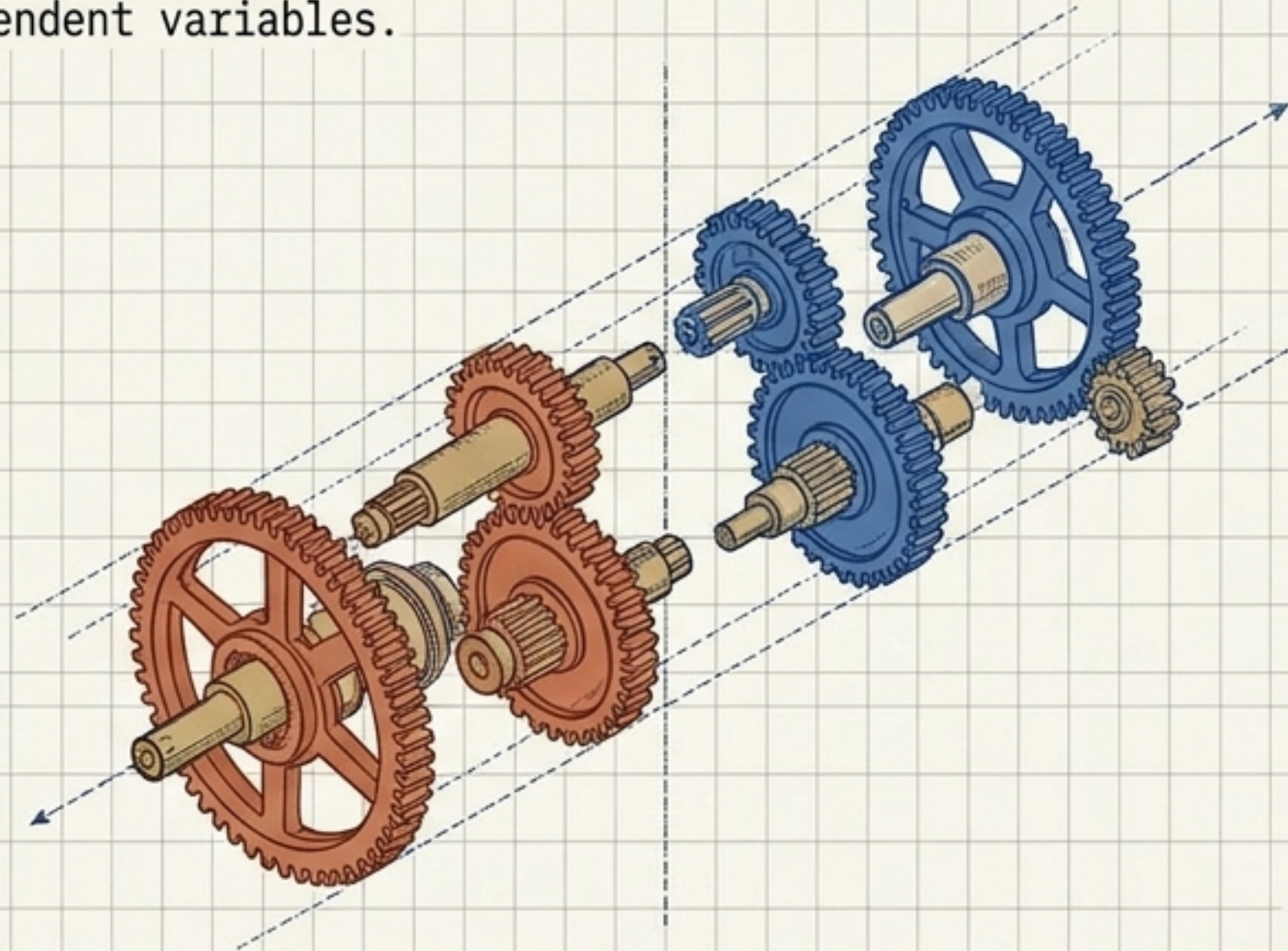
Description-Length Non-Identification Theorem

Fixing description length leaves distinguishing capacity completely free. No ordering of description lengths across two ontologies constrains the ordering of their distinguishing capacities.

- Magnitude does not determine Topology.
- They are fully independent variables.



Description Length (L_T)

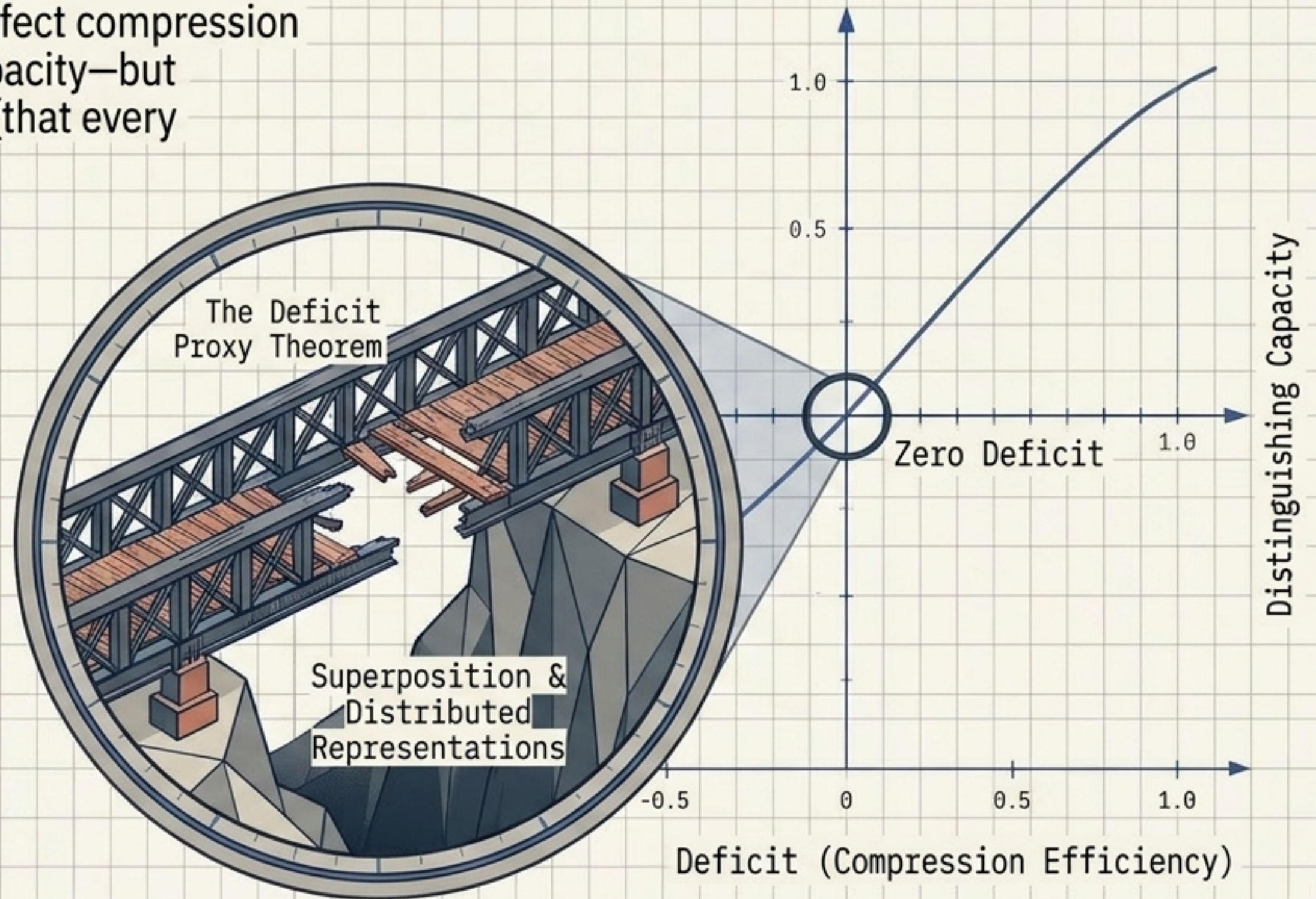


Distinguishing Capacity (D_T)

The Fatal Flaw of the Faithful Encoding Caveat

The Deficit Proxy Theorem claims perfect compression guarantees perfect distinguishing capacity—but only if we assume **Faithful Encoding** (that every distinction costs at least one bit).

In modern ML architectures relying on distributed representation and superposition, this assumption fails entirely. Faithful encoding does not hold, meaning the proxy bridge collapses everywhere along the working range.



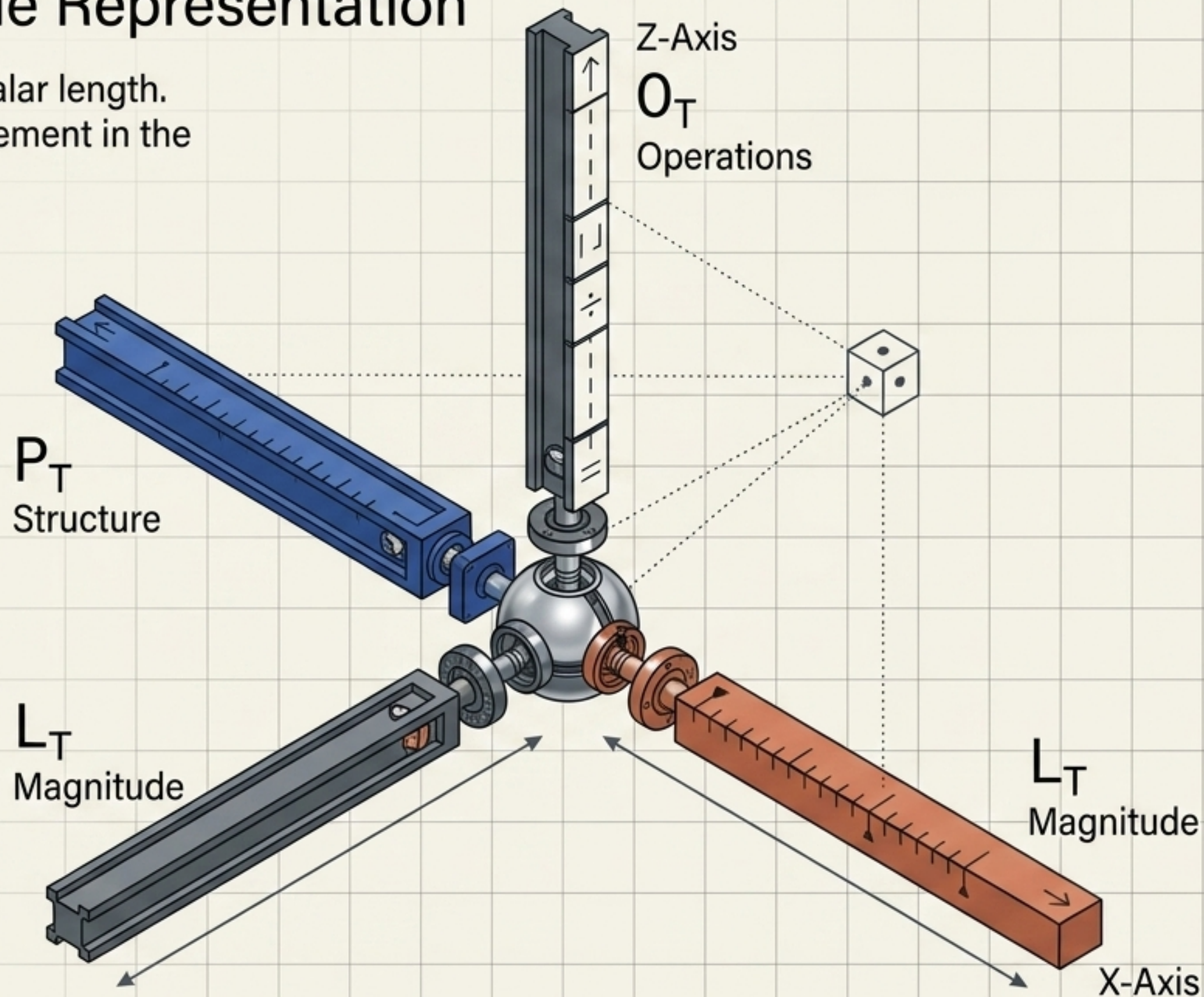
The Three Coordinates of True Representation

True understanding cannot be measured by a scalar length.
Improving one coordinate does not imply improvement in the others.

- **Magnitude** (L_T) - The visible size of the description.
- **Structure** (P_T) - The topology of what is kept distinct vs. collapsed.
- **Operations** (O_T) - What can be computed natively over the encoded representation.

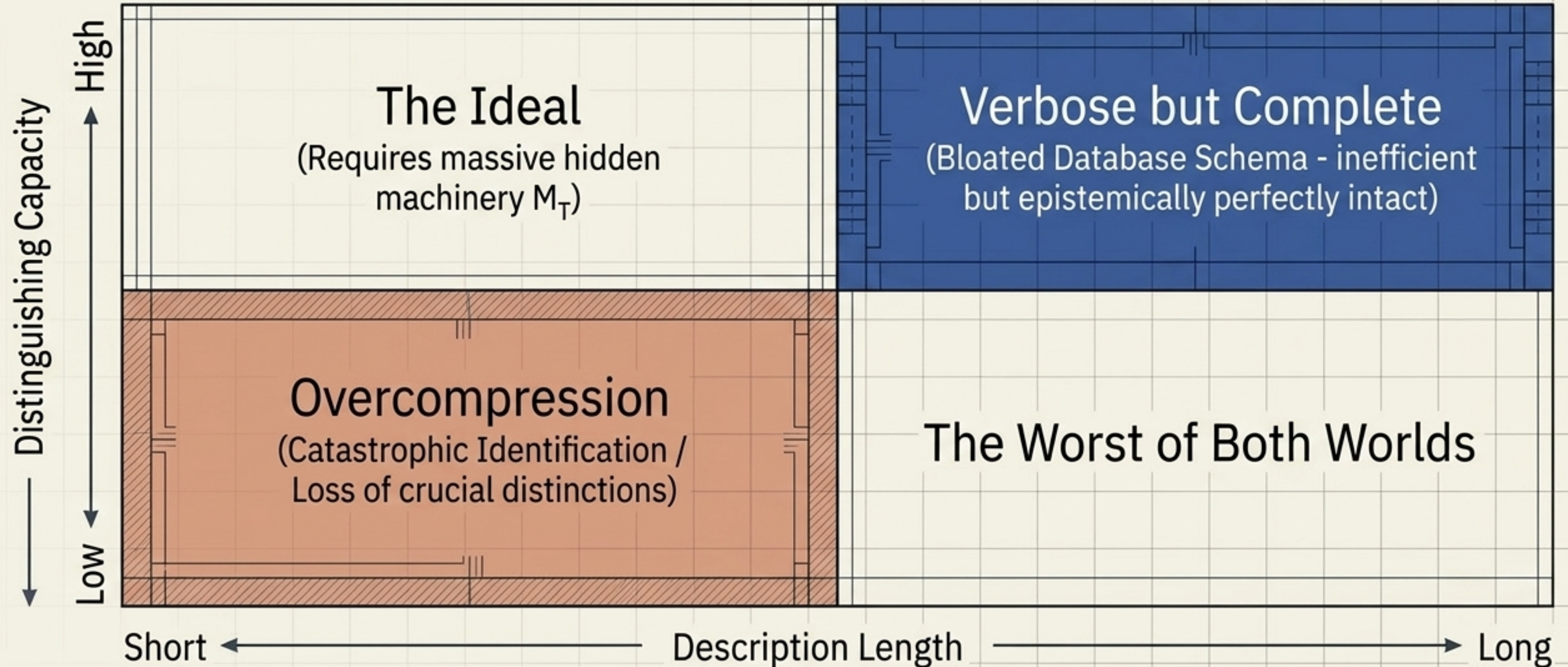
$$L_T \not\Rightarrow P_T$$

$$L_T \not\Rightarrow O_T$$



Navigating the Epistemic Topologies of Compression

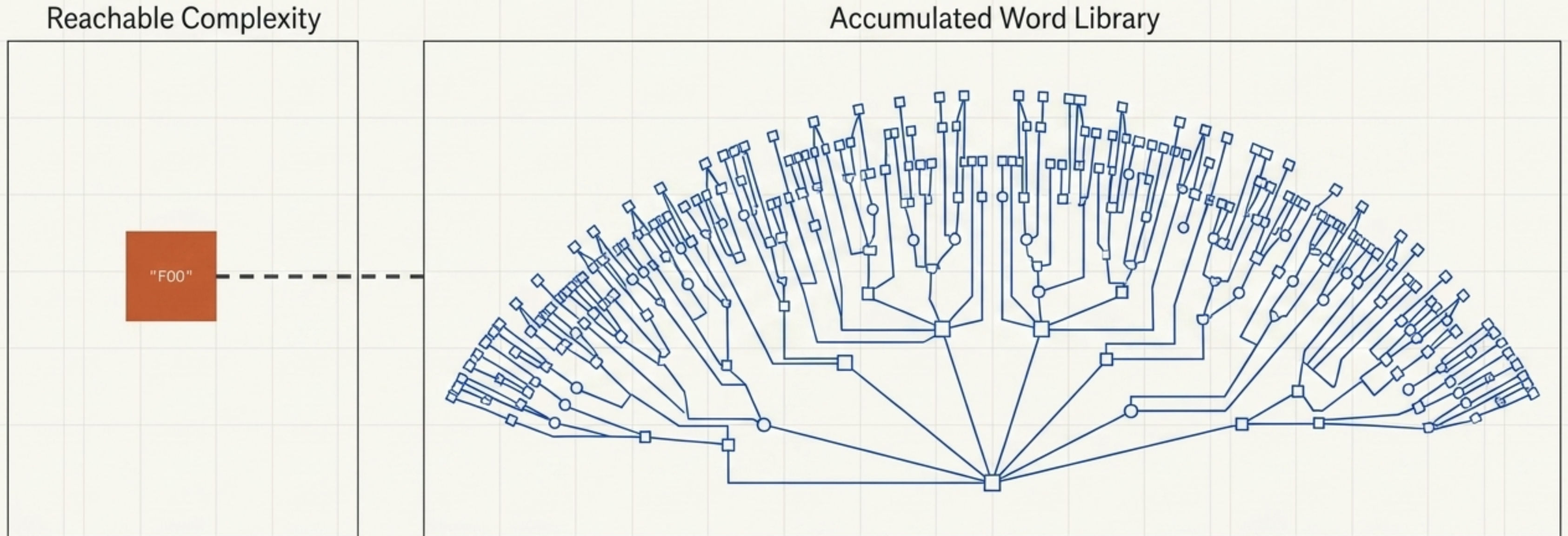
Compression progress does not automatically push a system to the top left.
The drive for brevity often pulls systems directly into the Overcompression zone.



Case Study I: Reachable Complexity in Forth

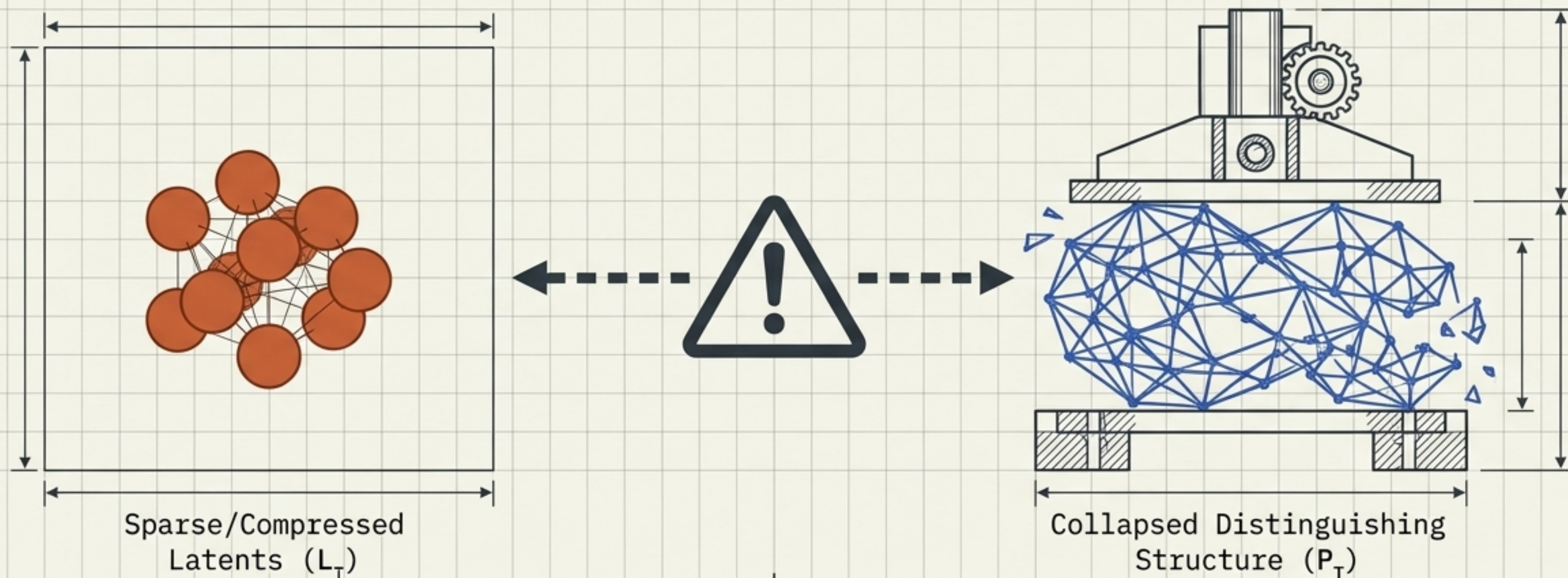
Forth code is incredibly short. This is not because it achieves optimal Kolmogorov complexity ($L^*(x)$), but because it relies on a prior achievement: an accumulated library of admissible operations.

The Insight: Its terseness is a triumph of machinery displacement, not a measure of absolute informational floors.



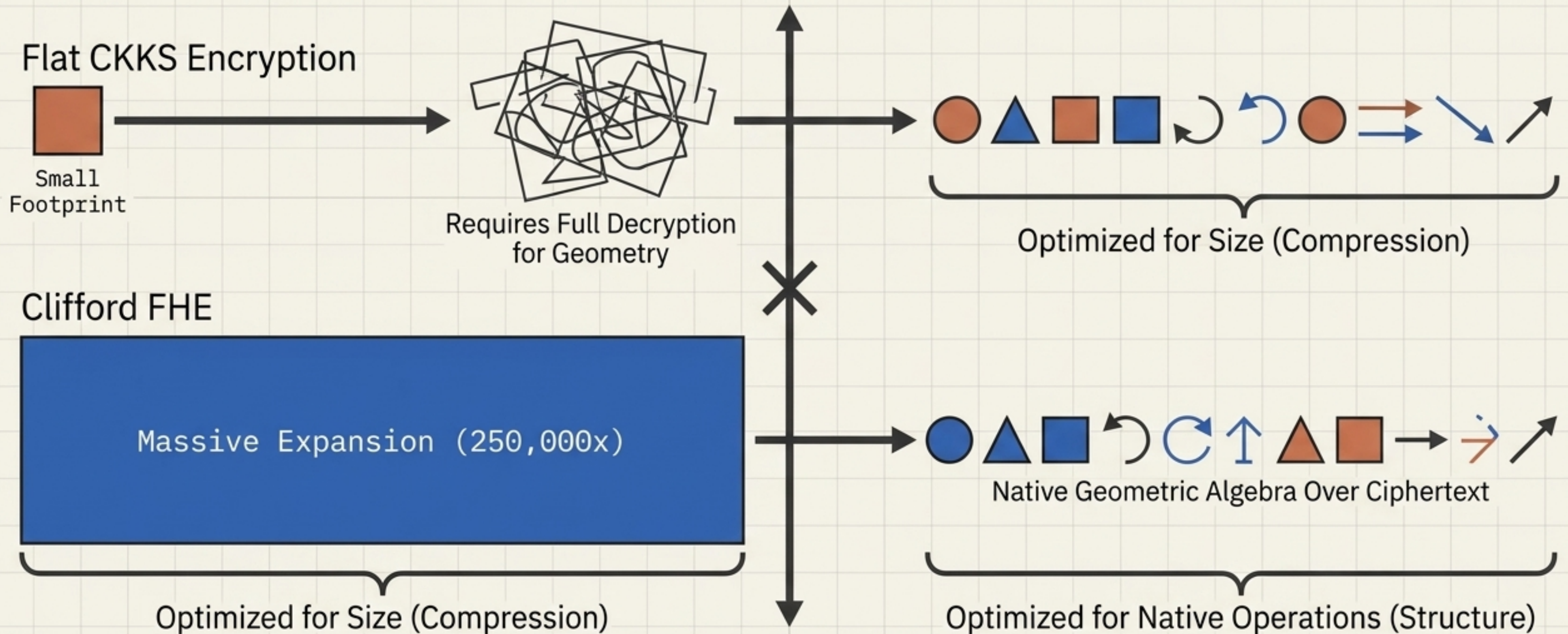
Case Study II: The Trap of Sparse Latent Ontologies

Modern interpretability often treats the discovery of a smaller, cleaner set of feature directions as evidence of genuine ontology. Because $L_T \neq P_T$, treating smaller feature directions as evidence of truth is mathematically ungrounded. We optimize for shortness and merely assume it yields understanding.



Case Study III: Preserving Structure Over Size

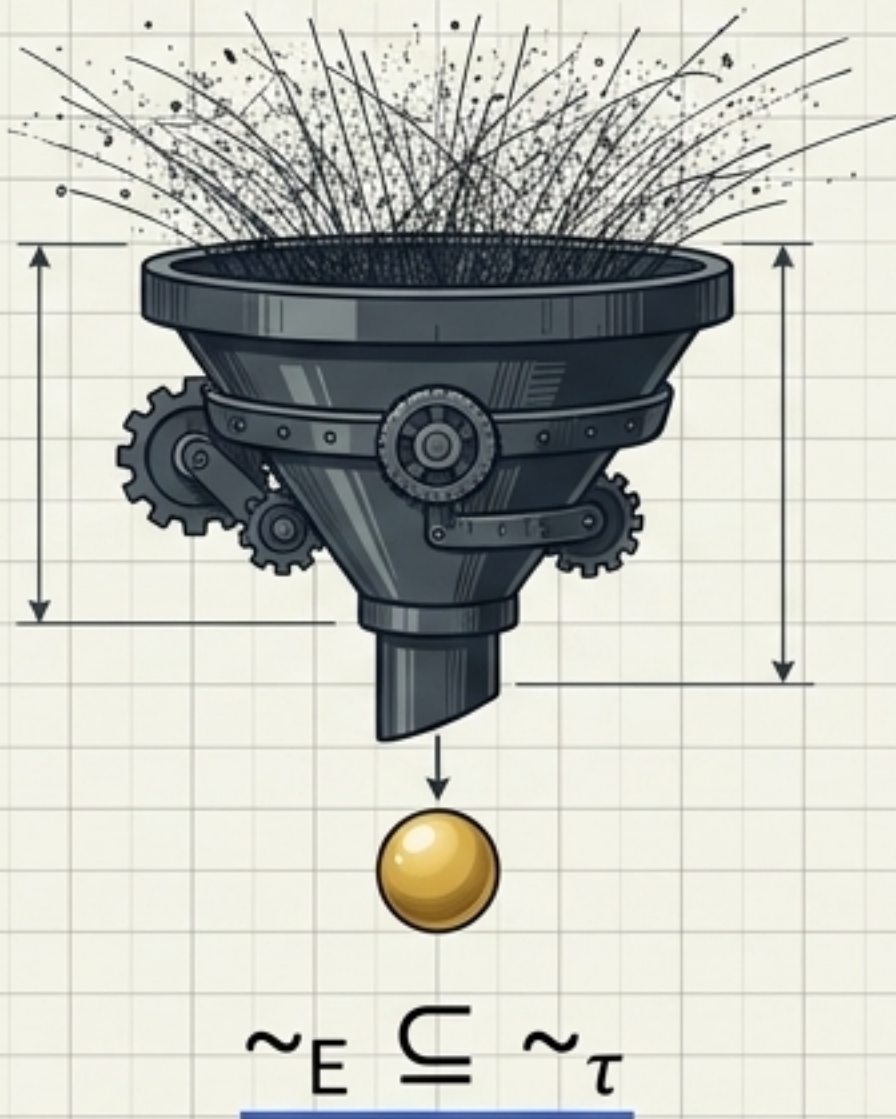
A representation's stored values don't dictate its utility—what matters is what you can compute over it (O_T). Clifford FHE abandons compression entirely to preserve an algebra of operations.



Task-Relative Equivalence vs. Overcompression

Not all collapsed distinctions are bad. Compression is only meaningful relative to a prior decision about what must survive. True understanding requires overcompression relative to the domain, but perfect preservation relative to the target.

Sufficient Statistic

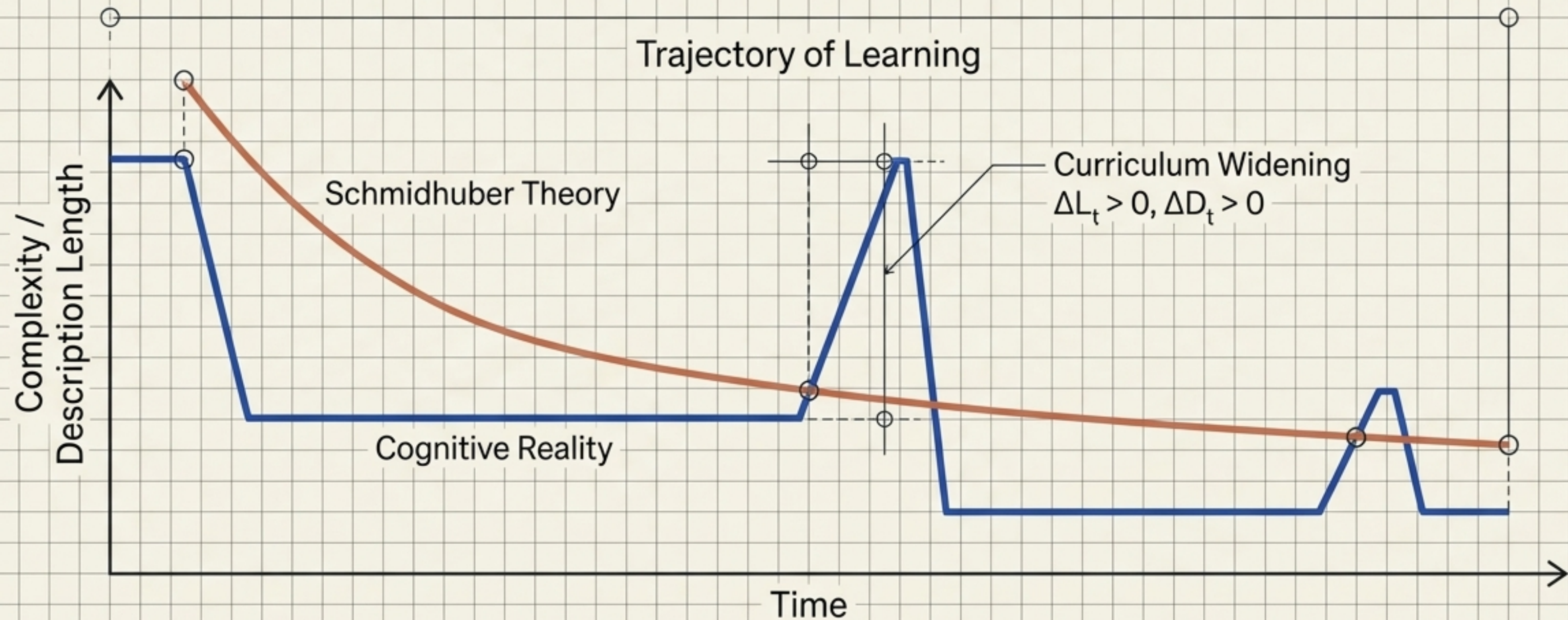


Overcompression



The Pedagogy Problem: Why Curricula Must Widen

If continuous compression equaled learning, we would never seek complexity. Productive problem-solving requires deliberate detours through greater complexity to map new distinctions before they can be compressed.

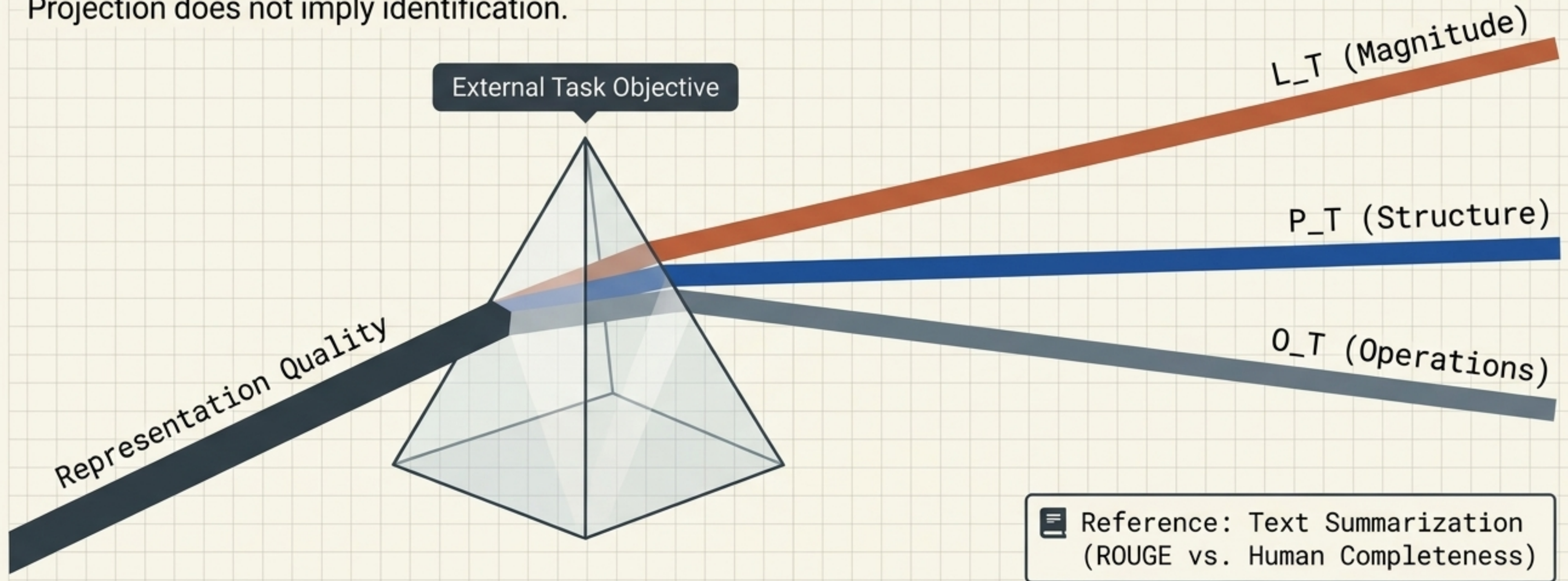


The Illusion of the Scalar Metric

Representational dominance is not a total order (Proposition 5).

There is no universal scalar function $U(T)$ derivable purely from description length.

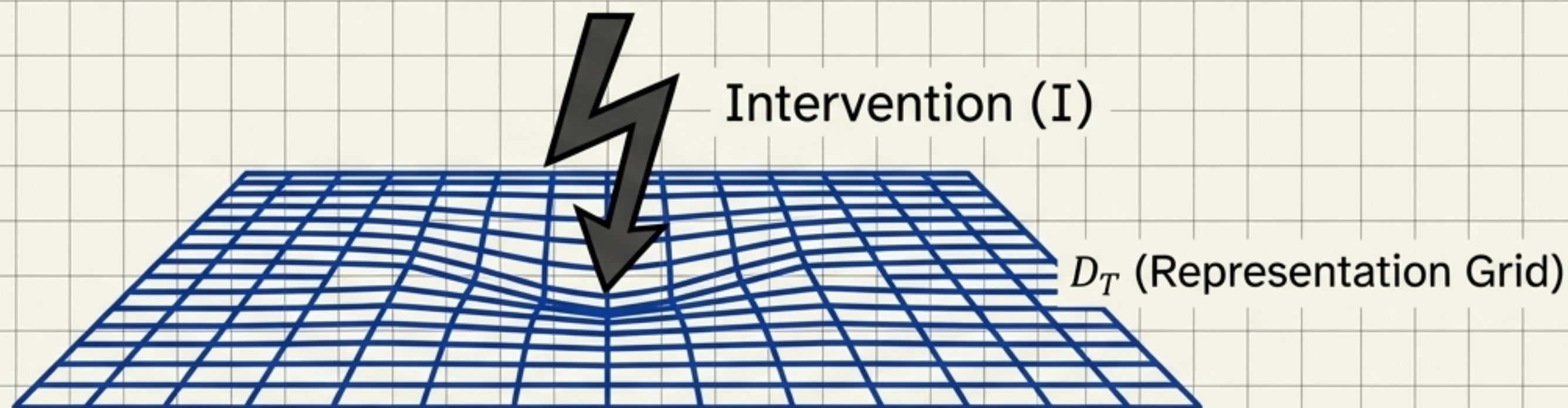
You cannot rank representations without external values dictating how economy, structure, and operation trade off. Projection does not imply identification.



The Actionable Standard: Understanding Under Intervention

How do we measure understanding if not by length?

Measure Distinguishing Capacity under deliberate intervention.



$$R_T(x, I) = D_T(x \mid I) - D_T(x)$$

Intervention Robustness (R_T)

Don't ask how many bits encode the resting state; ask which relational distinctions survive an admissible transformation of the domain.

Moving Beyond the Fallacy: New Best Practices



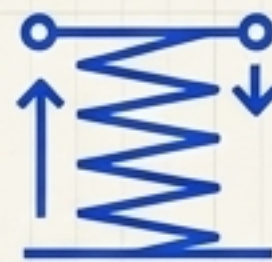
1. Audit the Machinery

Never report **visible length** ($L_{visible}$) without accounting for the background decoder or accumulated library (M_T).



2. Check for Superposition

Stop using size as a proxy for **distinguishing power** in neural latents. Where **faithful encoding fails**, the proxy theorem collapses.



3. Perturb, Don't Just Compress

Evaluate representational success by its **capacity** to **separate necessary objects** under **perturbation** (R_T), not by how few bits it occupies at rest.